JUSTIN CLANCY

Data Scientist | Astrophysicist

justinpclancy@gmail.com   +61 427 454 239   justin-clancy   github.com/justinc97

PROFESSIONAL SUMMARY

Data scientist with 4+ years of experience developing processing and analysis pipelines, and statistical models for large-scale datasets. Passionate about exploring and applying data, technology and critical thinking to drive change in independent or community environments. Strong experience in Python, data analysis, modelling, and high performance computing (HPC). Strong communicator capable of translating complex technical concepts to diverse audiences. Highly motivated by social impact and engaging with the community.

TECHNICAL SKILLS

Programming: Python, R, SQL, Bash, Slurm, HTML
Data Science & Machine Learning: scikit-learn, Pandas, SciPy, NumPy
Statistical Methods: Bayesian inference, time series analysis, regression, bootstrap error analysis, Fourier analysis
Data Engineering: ETL pipelines, data modelling, Apache Spark, PostgreSQL, MySQL, data validation
Data Visualization: Matplotlib, Plotly, Seaborn, Tableau, interactive dashboards, Flask
DevOps & Tools: Git, Docker, Jupyter, Linux/Unix, HPC clusters & distributed data processing, version control, LaTeX, MPI
Soft Skills: Cross-functional collaboration, technical writing, versatile communication, mentoring, project management

PROFESSIONAL EXPERIENCE

LLM Training Specialist — *Oct 2025 – Present*
Mercor Intelligence — Independent Consultant

- Developing prompts and tasks in scientific domains, formulating complex multi-step problems designed to challenge model reasoning capabilities.
- Documenting comprehensive solutions and methods to enhance LLM training datasets in expert fields such as physics, astrophysics, and mathematics.

Research Data Scientist (Contributions during postgraduate studies) — *Jan 2022 – Present*
Simons Observatory Collaboration — Melbourne, VIC, Australia

- Handled large-scale datasets with Python, Julia, and high performance computing (HPC) tools for optimization and parallelization across computing environments
- Constructed pipelines for simulation, data processing, and analysis.
- Led technical problem solving across teams, managing multiple tasks and projects while serving as a data expert within a multidisciplinary collaboration.
- Communicated findings at domestic and international conferences, and to the general public.

  Projects:

  A Pipeline to Detect Rapid Transients in Time-Ordered Data

  – Developed and tested a pipeline to process large-scale, high-frequency time series data from the Simons Observatory large aperture, millimetre-wave telescope, to detect rapid astrophysical transients.
  – Uses a matched-filtering approach, with density-based clustering and Fourier analysis to maximise and isolate desired signals.
  – The pipeline was tested on simulated time series data, achieving $\geq 90\%$ detection efficiency for flaring events as short as 0.5 s.
  – The pipeline is currently being integrated into the full Simons Observatory software suite for real-time detection: Simons Observatory Time Ordered Data Library
  – Submitted to the Open Journal of Astrophysics: NASA ADS

  Assessing the Observational Impacts of the Relativistic Sunyaev-Zel'dovich Effect on Cluster Catalogue Creation

  – Generated large-scale simulated maps of the sky including various theoretical and observational effects.

- Constructed catalogues from these maps using multi-frequency matched filtering.
- Compared the various effects through statistical comparison of correlated and uncorrelated signals.

Polarization fraction of Planck Galactic cold clumps and forecasts for the Simons Observatory

- Processed and analysed large datasets from the *Planck* satellite.
- Performed a stacking analysis with bootstrap error calculation to measure the polarization fraction from protostellar cores throughout the Milky Way.
- Developed a mask generator for CMB experiments using publicly available catalogues of objects: Github
- Published in Monthly Notices of the Royal Astronomical Society: NASA ADS

Undergraduate Teaching — *Mar 2020 – October 2025*
The University of Melbourne — Melbourne, VIC, Australia

- Taught and supported over 400 undergraduate students in physics courses.
- Prepared course content and tools, enabling smooth delivery and accuracy.
- Led physics labs, demonstrating data acquisition, processing and analysis.

EDUCATION

Doctor of Philosophy (PhD) - Science — *2022 – 2025*
The University of Melbourne — Melbourne, VIC, Australia

Master of Science (Physics) – With Distinction — *2020 – 2021*
The University of Melbourne — Melbourne, VIC, Australia

Bachelor of Science (Major in Physics) — *2016 – 2018*
The University of Melbourne — Melbourne, VIC, Australia

AWARDS & SCHOLARSHIPS

- **Dr Albert Shimmins Postgraduate Writing-Up Award**, The University of Melbourne (2025)
- **Laby PhD Travelling Scholarship**, The University of Melbourne (2024)
- **Research Training Program Scholarship**, The University of Melbourne & The Australian Government (2022–2025)
- **Melbourne Research Scholarship**, The University of Melbourne (2022)
- **Klein Prize in Experimental Physics**, School of Physics, The University of Melbourne (2022)

References

Available upon request.